

Question Paper

Exam Date & Time: 28-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Engineering Chemistry [CHM-1071]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the construction and working of calomel electrode as both cathode and anode. Mention any two disadvantages of it. (4)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) i) Calculate EMF of the zinc-silver cell at 25 °C with $[Zn^{2+}] = 3\text{ M}$ and $[Ag^+] = 15\text{ M}$. Write the cell representation and electrode reactions. (3 MARKS) (4)
ii) Justify: Aqueous electrolytes are not preferred in lithium ion batteries.(1 MARKS)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Give reasons for the following: (2)
i) Ambient air cannot be used in alkaline fuel cell.
ii) Pure chromium is not used as anode during chromium plating.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) I) Define Beer-Lambert's law and write the mathematical expression. Mention any two of its limitations. (2 Marks) (4)
A) ii) With relevant graphical representation explain the conductometric titration of strong acid vs strong base. (2 Marks)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) i) What is priming? How can it be avoided? (2 Marks) (4)
ii) Describe the carbonate and phosphate conditioning of water to overcome the boiler feed problem. (2 Marks)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) Distinguish between (2)
i) Scale and sludge
ii) Temporary and permanent hardness of water

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3) i) Discuss the caustic embrittlement of mild steel with necessary reactions. (3 Marks) (4)
ii) Give an example for oxidation corrosion.(1 Mark)

A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) i) How does hydrogen overvoltage affect the rate of corrosion? **(1 Mark)** (4)
ii) Explain in detail the anodic protection method of corrosion control. **(3 Marks)**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- C) Identify and explain the following forms of corrosion: (2)
i) Bolt and nut made from different metals when in contact.
ii) Presence of small amount of iron in aluminium.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) i) Explain any four factors affecting the glass transition temperature. **(2 Marks)** (4)
ii) Discuss the classification of thermotropic liquid crystals based on shape. **(2 Marks)**

A)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- B) i) Write differences between thermoplastic and thermosetting polymers. **(1 Mark)** (4)
ii) The polymer sample has the following composition.

Degree of polymerization	250	350	450	550
% of composition	10	20	30	40

If the number average molecular weight of the sample is 11800. Calculate the molecular weight of the monomer, weight average molecular weight and PDI of the polymer sample. **(3 Marks)**

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) With a neat diagram explain the PVD process. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5) i) Define concentration polarization. **(1 mark)** (4)
ii) The EMF of a cell which is independent of the concentration of the KCl solution $\text{Pb} / \text{PbCl}_2(\text{s}) // \text{KCl soln} / \text{Hg}_2\text{Cl}_2 / \text{Hg}$ is 0.855 V at 298 K and the temperature coefficient is $-3.38 \times 10^{-4} \text{ VK}^{-1}$.
A) Calculate the change in free energy, enthalpy and entropy changes at 298 K. **(3 Marks)**

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) i) Discuss the particle reinforced composites. **(2 Marks)** (4)
ii) Define hydrogen bonding. Justify p - nitrophenol has higher boiling point than o - nitrophenol. **(2 Marks)**

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the role of anodic inhibitor in corrosion control. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CHM-1071:Engineering Chemistry

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	4

3	3	CO 3	12
4	4	CO 4	6
5	5	CO 5	14
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CHM-1071:Engineering Chemistry

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	12	38(76%)
3	Application	3	12(24%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)